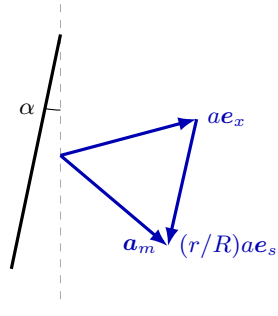


$$Ma = T_s[-\sin(\alpha+\theta)] + f - Mg\sin\theta$$



$$a\cos(\alpha+\theta) = g\sin\alpha$$